MAST20026 REAL ANALYSIS

The Definition of Differentiation - Solutions

9B

A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

1. **A Differentiable Piecewise Function**

   Noting that $f(1) = 1$, we compute that

   $$\lim_{x\to 1^-} \frac{f(x) - f(1)}{x - 1} = k \quad \text{and} \quad \lim_{x\to 1^+} \frac{f(x) - f(1)}{x - 1} = 3.$$

   Therefore f differentiable at $c = 1$ if and only if $k = 3$.

2. **A Flawed Proof**

   Differentiability implies continuity. However, f is not continuous at $c = 0$, as $\lim_{x\to 0^+} f(x) = 0 \neq 4 = f(0)$. Therefore f is not differentiable at $c = 0$.

   There is a flaw in the proof here:

   When $x > 0$ we have $f'(x) = 2x + 2$. Therefore $f'(0) = 2$. Therefore

   $$\lim_{x\to 0^+} \frac{f(x) - f(0)}{x} = 2$$

   If we take the time to verify

   $$\lim_{x\to 0^+} \frac{f(x) - f(0)}{x} = 2$$

   we will notice a problem. f is not continuous at 0, we cannot substitute $x = 0$ as part of our work.

3. **Derivatives of Increasing Functions**

   Since f is differentiable on $[a, b]$, for all $c \in [a, b]$ there exists $L \in \mathbb{R}$ so that

   $$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = L.$$

   Examining the left and right side limits, for all $c \in (a, b)$ we have

   $$\lim_{x\to c^-} \frac{f(x) - f(c)}{x - c} = L = \lim_{x\to c^+} \frac{f(x) - f(c)}{x - c}.$$

   Approaching from the right, we have $x > c$. Since f is monotone increasing, we have $f(x) \geq f(c)$. Therefore

   $$\frac{f(x) - f(c)}{x - c} \geq 0$$

   Therefore

   $$L = \lim_{x\to c^+} \frac{f(x) - f(c)}{x - c} \geq 0.$$

   By a similar argument, $f'(a) \geq 0$ and $f'(b) \geq 0$. Therefore $f'(c) \geq 0$ for all $c \in [a, b]$.

4. **Locally Non-Constant**

Let f be differentiable on $[a, b]$ and let $c \in (a, b)$.

(a) We prove the contrapositive. Assume for all $\delta > 0$ the statement

$$f(x) \neq f(c) \textit{ for all } x \in (c-\delta, c+\delta) \textit{ with } x \neq c.$$

is false. Therefore for all $\delta > 0$ there exists $x \in (c-\delta, c+\delta)$ so that $x \neq c$ and $f(x) = f(c)$. We prove $f'(c) = 0$.

Since f is differentiable on $[a, b]$, there exists $L \in \mathbb{R}$ so that

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = L.$$

Therefore for every $\epsilon > 0$ there exists $\delta > 0$ so that $\left|\frac{f(x)-f(c)}{x-c} - L\right| < \epsilon$ whenever $0 < |x - c| < \delta$. By hypothesis, for every such δ there exists $z \in (c-\delta, c+\delta)$ so that $f(z) = f(c)$. And so, for every $\epsilon > 0$ there exists $z \in (c-\delta, c+\delta)$ so that

$$\epsilon > \left|\frac{f(z) - f(c)}{z - c} - L\right| = |-L| \geq 0.$$

Therefore we have

$$0 \leq |-L| < \epsilon$$

for every $\epsilon > 0$. And so we conclude $L = 0$.

(b) No. Consider the function $f(x) = x^3$. This function is strictly increasing, so for all $x \neq 0$, $f(x) \neq f(0) = 0$. But, $f'(0) = 0$

5. **A Discontinuous Derivative**

(a) Yes. To show this, you can use the fact that $|\sin(1/x)| \leq 1$ for $x \neq 0$, and hence

$$-x^2 \leq f(x) \leq x^2.$$

Since $\lim_{x \to 0} x^2 = 0$, the Squeeze Theorem for functions immediately gives

$$\lim_{x \to 0} f(x) = 0 = f(0).$$

(b) Using the usual rules, we find that, for $c \neq 0$,

$$f'(c) = 2c \sin \frac{1}{c} - \cos \frac{1}{c}.$$

(c)

$$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} = 0$$

Therefore

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = 0.$$

(d) Let

$$g(x) = \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Since 0 is a limit point of the domain, to prove g is not continuous at 0 it suffices to prove

$$\lim_{x \to 0} g(x) \neq g(0)$$

In fact, $g(x)$ does not converge as x approaches 0. (For all $\delta > 0$, there is an $x_1 \in (0, \delta)$ such that $\cos \frac{1}{x_1} = 1$, and also $x_2 \in (0, \delta)$ such that $\cos \frac{1}{x_2} = -1$.)

Therefore $\lim_{x \to 0} g(x) \neq g(0)$, and g is not continuous at 0.